

# Technical Appendix: Sampling-boundary distortion in Sharpe ratio evidence from financial panels

## 1 Purpose

This appendix provides derivations, calibration checks, robustness tables, and reproducibility details omitted from the main manuscript for brevity. It supports the proof, calibration, and robustness record; it is not an independent empirical claim.

## 2 Public-Panel Momentum Implementation

The public French panel implementation uses a skipped-momentum signal as the running row-level illustration. The sampling-boundary diagnostic does not depend on this particular signal, but the formulas below define the exact public-panel construction used in the reported tables. For lookback  $L$  and skip  $q$ , the skipped compounded signal is

$$s_{t,i}^{(L,q)} = \prod_{j=t-L-q+1}^{t-q} (1 + r_{j,i}) - 1. \quad (1)$$

With  $q \geq 1$ , the signal excludes the outcome return and is executable at date  $t$  when all signals in the available universe  $A_t$  are known before the trading decision. The signed one-period row return is

$$y_{t+1,i}^{(L,q)} = \text{sign}(s_{t,i}^{(L,q)}) r_{t+1,i}, \quad (2)$$

with  $\text{sign}(0) = 0$ ; zero-signal and missing-signal rows receive zero economic weight. The confidence score  $c_{t,i}^{(L,q)}$  is the within-date percentile rank of  $|s_{t,i}^{(L,q)}|$ , and the selected set is

$$\mathcal{S}_t(L, q, \tau) = \{i : c_{t,i}^{(L,q)} \geq \tau\}.$$

The percentile score is a baseline screening rule, not an optimal weighting theorem. The replication files record tie handling, one-name dates, selected-count granularity, and source-specific frequency and annualization metadata.

### 3 Mean-Return Design-Effect Proof

For a selected row model  $y_{t,i} = \mu + \lambda_t + \epsilon_{t,i}$  with constant selected count  $\bar{n}$ , same-date intraclass correlation  $\rho$ , and independent dates, the row-pooled mean equals the average of date means:

$$\hat{\mu} = \frac{1}{T\bar{n}} \sum_{t=1}^T \sum_{i=1}^{\bar{n}} y_{t,i} = \frac{1}{T} \sum_{t=1}^T \bar{y}_t.$$

The row-naive variance report is  $\sigma^2/(T\bar{n})$ . The true one-date variance is

$$\text{Var}(\bar{y}_t) = \frac{1}{\bar{n}^2} \{ \bar{n}\sigma^2 + \bar{n}(\bar{n} - 1)\rho\sigma^2 \} = \frac{\sigma^2}{\bar{n}} \{ 1 + (\bar{n} - 1)\rho \}.$$

Thus

$$\frac{se_{date,true}}{se_{row,naive}} = \sqrt{1 + (\bar{n} - 1)\rho}.$$

This is the Moulton-like design effect used as the mean-return special case in the main text. It is a closed-form special case, not the full Sharpe-ratio distortion result.

### 4 Sharpe UVIF Derivation

These calculations supply the algebra behind the Sharpe UVIF decomposition in the main manuscript. Let  $R_t$  be the date-level portfolio excess return and  $G_t = (R_t, R_t^2)'$ . The annualized Sharpe ratio is the smooth functional

$$f(\mu, m_2) = \sqrt{A_f} \frac{\mu}{(m_2 - \mu^2)^{1/2}}, \quad d_R = \nabla f(\mu_R, m_{2,R}).$$

The annualization factor  $A_f$  is source-specific: the public configuration uses 252 for daily sources and 12 for monthly sources. On sets where  $v_R = m_{2,R} - \mu_R^2 > 0$ , the first-order expansion is

$$\widehat{\text{SR}}_R - \text{SR}_R = d'_R(\hat{\theta}_R - \theta_R) + o_p(T_R^{-1/2}), \quad \theta_R = \mathbb{E}[G_t].$$

With long-run covariance  $\Omega_R = \Gamma_0^R + \sum_{k \geq 1} (\Gamma_k^R + \Gamma_k^{R'})$ ,

$$\text{Var}_{LR}(\widehat{\text{SR}}_R) = T_R^{-1} d'_R \Omega_R d_R + o(T_R^{-1}).$$

The row-pooled IID calculation applies the same delta method to  $\bar{n}T_R$  row observations  $H_{t,i} = (Y_{t,i}, Y_{t,i}^2)'$ :

$$\text{Var}_{IID,row}(\widehat{\text{SR}}_{row}) = (\bar{n}T_R)^{-1} d'_Y \Psi_Y d_Y + o((\bar{n}T_R)^{-1}).$$

Taking the ratio gives the Sharpe UVIF:

$$\text{UVIF}_{\text{SR}} = \frac{\bar{n} d'_R \Omega_R d_R}{d'_Y \Psi_Y d_Y} = \frac{\bar{n} d'_R \Gamma_0^R d_R}{d'_Y \Psi_Y d_Y} \times \frac{d'_R \Omega_R d_R}{d'_R \Gamma_0^R d_R}.$$

The first term is the cross-sectional component and the second term is the serial long-run factor. Componentwise,

$$d'_R \Omega_R d_R = \frac{A_f}{v_R^3} \left[ m_{2,R}^2 \Omega_{11} - \frac{\mu_R m_{2,R}}{2} (\Omega_{12} + \Omega_{21}) + \frac{\mu_R^2}{4} \Omega_{22} \right].$$

The  $\Omega_{22}$  term contains long-run covariances of squared returns and therefore brings fourth moments into Sharpe-ratio variance.

## 5 Optional Order-Size Sensitivity

The public sorted-portfolio examples in the main manuscript do not contain average daily volume or order-size information, so they do not support an execution-scale claim. For traded-security panels where those inputs are available, a square-root impact diagnostic can be added as an optional sensitivity extension.

**Proposition 1** (Square-root order-size sensitivity bound). *Suppose a traded panel has row-pooled gross mean  $\bar{r}_{\text{gross}}$ , row-pooled return standard deviation  $\sigma_{\text{row}}$ , strictly positive average turnover  $\widehat{\text{TO}}$ ,  $T$  trading dates, square-root impact cost  $\lambda \widehat{\text{TO}} \sqrt{A/V}$ , and a dependence standard-error inflation factor  $\mathcal{M} \geq 1$  relative to row-pooled inference. For a one-sided level- $\alpha$  sensitivity margin, any statistically screened asset scale  $A^*$  must satisfy*

$$A^* \leq V \left( \frac{\bar{r}_{\text{gross}} - z_\alpha \mathcal{M} \sigma_{\text{row}} / \sqrt{T}}{\lambda \widehat{\text{TO}}} \right)^2 \quad (3)$$

when the numerator is positive; otherwise no positive scale is screened by the margin.

For a square-root impact diagnostic, the dependence-adjusted one-sided lower confidence margin for gross mean return is

$$\bar{r}_{\text{gross}} - z_\alpha \mathcal{M} \sigma_{\text{row}} / \sqrt{T},$$

where  $\sigma_{\text{row}}$  is the standard deviation of row-pooled returns before date aggregation or UVIF adjustment. Square-root impact subtracts  $\lambda \widehat{\text{TO}} \sqrt{A/V}$ . Requiring the net lower margin to be nonnegative gives

$$\lambda \widehat{\text{TO}} \sqrt{A/V} \leq \bar{r}_{\text{gross}} - z_\alpha \mathcal{M} \sigma_{\text{row}} / \sqrt{T}.$$

Rearranging yields the sensitivity bound in Proposition 1. If the volatility input is already the date-level portfolio volatility, the date boundary has already embedded the cross-sectional UVIF component, so the cross-sectional part of  $\mathcal{M}$  must not be multiplied in again.

## 6 Higher-Moment UVIF and ERIR Details

For  $\theta^{(4)} = (\mu, m_2, m_3, m_4)'$ , define skewness and kurtosis from the central moments implied by  $\theta^{(4)}$ . Any adjusted Sharpe functional  $h(\theta^{(4)})$  has a delta-method variance  $\nabla h' \Omega^{(4)} \nabla h$ , where  $\Omega^{(4)}$  is the long-run covariance matrix of  $(R_t, R_t^2, R_t^3, R_t^4)'$ . The higher-moment UVIF is the corresponding ratio of date-level long-run variance to row-pooled IID variance. It is a diagnostic extension, not an additional empirical rejection rule.

The Effective Row Information Retention (ERIR) diagnostic is a design-effect retention calculation, not a hypothesis test or a formal information claim. Under the same equicorrelated approximation used for the mean-return design effect,

$$DE_{\mu, xs} = 1 + (\bar{n} - 1)\rho, \quad \text{ERIR} = \{1 + (\bar{n} - 1)\rho\}^{-1}.$$

Thus an ERIR of 0.111 means that roughly  $1 - 0.111 = 0.889$ , or 89 percent, of nominal row-level information is redundant under this diagnostic approximation. Confirmatory inference remains based on date-level HAC-delta and dependent-bootstrap procedures. When the descriptive equicorrelation estimate is non-positive, the unclipped ERIR can exceed one and the corresponding UVIF can fall below one; the main manuscript displays a conservative one-sided UVIF floor at one and reports the unclipped diagnostics separately.

## 7 Synthetic Positive Control

This section gives the appendix version of the known-edge calibration check. The synthetic panel is not empirical evidence for a public strategy; it verifies that the pre-specified reporting rule can accept a declared positive signal when the data-generating process contains one. The generated panel has  $T = 5000$  dates and  $N = 100$  names. Signed returns include daily alpha 0.12/252, idiosyncratic volatility 0.01, a common loading of 0.50, AR(1) common-factor coefficient 0.10, signal-stay probability 0.98, and seed 777. The displayed run is one representative known-edge calibration with 5000 bootstrap draws and 10000 same-date permutations; costs use the same 5 bps turnover-stress convention as the public tables. The replication screen below uses fewer resampling draws and is

reported only as a power sanity check. The data-generating process is

$$f_t = 0.10f_{t-1} + u_t, \quad \epsilon_{t,i} \sim N(0, 0.01^2),$$

$$z_{t,i} \in \{-1, 1\}, \quad \Pr(z_{t,i} = z_{t-1,i}) = 0.98,$$

$$r_{t+1,i} = z_{t,i}(0.12/252) + 0.50f_t + \epsilon_{t,i}, \quad s_{t,i} = z_{t,i}.$$

The calibration then applies the same signed-row construction, date aggregation, HAC-delta inference, Romano-Wolf menu adjustment, same-date permutation, and 5 bps cost stress used in the public tables.

Table 1. Synthetic positive-control evaluation.

| Panel             | $\bar{\rho}$ | SE infl. | SR    | HAC $p^+$ | RW $p$   | Perm. $p^+$ |
|-------------------|--------------|----------|-------|-----------|----------|-------------|
| Synthetic control | 0.202        | 4.58     | 1.744 | 7.67e-13  | 2.00e-04 | 1.00e-04    |

Table 2. Positive-control replication screen.

| Reps | Boot | Perms | Block pass | HAC pass | RW pass | Perm. pass | Cost pass | All pass |
|------|------|-------|------------|----------|---------|------------|-----------|----------|
| 100  | 499  | 199   | 100.0%     | 100.0%   | 100.0%  | 100.0%     | 100.0%    | 100.0%   |

*Note:* The replication screen uses fewer resampling draws than the detailed calibration row and is reported only as a power sanity check.

## 8 Full Empirical Tables

These tables are supporting diagnostics rather than primary evidence; the main text reports the compressed interpretation. The inferential tables in this appendix report only public sources that loaded and produced reproducible outputs. Failed source-access attempts remain in the generated provenance files and run metadata, but are not treated as empirical evidence. Panel candidates and pre-aggregated factor benchmarks are reported in separate groups because only the former provide row-level signals for same-date permutation and sampling-boundary diagnostics. For each empirical table group, the check is whether the result survives moving from rows to date returns, whether nearby HAC and dependent-bootstrap variants agree, and whether placebo or cost diagnostics overturn the simple Sharpe interpretation. Large disagreement across these tables is a robustness warning, not a new discovery channel.

This supplement reports loaded sources in separated groups. The loader provenance is retained in the generated metadata; it is not a comparable pass/fail table. The tables are supporting diagnostics rather than primary evidence; the main text reports the compressed interpretation.

Table 3. Source frequency and annualization metadata.

| Source                                        | type                 | period  | A_f     | dates |
|-----------------------------------------------|----------------------|---------|---------|-------|
| AQR Betting Against Beta, equity factors      | single_series_factor | daily   | 252.000 | 25281 |
| AQR HML Devil                                 | single_series_factor | daily   | 252.000 | 26590 |
| AQR Quality Minus Junk                        | single_series_factor | daily   | 252.000 | 17681 |
| AQR Value and Momentum Everywhere             | single_series_factor | monthly | 12.000  | 651   |
| French momentum deciles, rank-threshold panel | panel                | daily   | 252.000 | 26131 |
| French WML decile benchmark                   | panel                | daily   | 252.000 | 26131 |
| French 25 Size/BM, skipped-momentum panel     | panel                | daily   | 252.000 | 3477  |

Table 4. Panel candidates with row-level signal-return structure.

| Panel candidate                               | SR    | HAC p+   | Boot p+  | RW p     | Perm p   | Net SR | Full rule 5% | Applicable checks 5% |
|-----------------------------------------------|-------|----------|----------|----------|----------|--------|--------------|----------------------|
| French momentum deciles, rank-threshold panel | 0.447 | 7.54e-05 | 2.00e-04 | 2.00e-04 | N/A      | 0.447  | no           | yes                  |
| French WML decile benchmark                   | 0.497 | 5.53e-06 | 2.00e-04 | 2.00e-04 | 1.00e-04 | 0.497  | yes          | yes                  |
| French 25 Size/BM, skipped-momentum panel     | 0.058 | 0.412    | 0.422    | 0.480    | 0.943    | -0.050 | no           | no                   |

Table 5. Single-series factor benchmarks subject to time-series inference only.

| Single-series benchmark                  | SR    | HAC p+   | Boot p+  | RW p     | Scope            |
|------------------------------------------|-------|----------|----------|----------|------------------|
| AQR Betting Against Beta, equity factors | 0.741 | 3.08e-13 | 2.00e-04 | 2.00e-04 | time-series only |
| AQR Quality Minus Junk                   | 0.536 | 2.26e-05 | 2.00e-04 | 2.00e-04 | time-series only |
| AQR HML Devil                            | 0.336 | 0.004    | 0.002    | 0.002    | time-series only |
| AQR Value and Momentum Everywhere        | 0.572 | 1.62e-04 | 2.00e-04 | 2.00e-04 | time-series only |

*Note:* Time-series only means not eligible for same-date permutation, row-retention diagnostics, or the full panel decision rule.

Table 6. Canonical French momentum benchmark validation.

| Candidate                   | thr   | n     | start      | end        | panel SR | direct WML SR | scale |
|-----------------------------|-------|-------|------------|------------|----------|---------------|-------|
| French WML decile benchmark | 0.900 | 26131 | 1926-11-04 | 2026-04-30 | 0.497    | 0.497         | 0.500 |

Table 7. Comparison with common row and date-level inference choices for panel candidates.

| Panel candidate                               | row p+   | date-IID p+ | date-HAC p+ | block p+ | UVIF  |
|-----------------------------------------------|----------|-------------|-------------|----------|-------|
| French momentum deciles, rank-threshold panel | 1.70e-04 | 2.00e-04    | 7.54e-05    | 2.00e-04 | 1.000 |
| French WML decile benchmark                   | 4.85e-04 | 2.00e-04    | 5.53e-06    | 2.00e-04 | 1.000 |
| French 25 Size/BM, skipped-momentum panel     | 0.253    | 0.417       | 0.412       | 0.422    | 9.002 |

*Note:* The row-naïve column treats selected rows as independent. Date-IID, date-HAC, and block-bootstrap columns operate on the date-level portfolio return series.

Table 8. Selected-count and dependence diagnostics for panel candidates.

| Candidate                                     | status    | thr   | rows   | dates | avg names | rho    | SE infl. |
|-----------------------------------------------|-----------|-------|--------|-------|-----------|--------|----------|
| French momentum deciles, rank-threshold panel | Available | 0.700 | 104524 | 26131 | 4.000     | -0.138 | 0.766    |
| French WML decile benchmark                   | Available | 0.900 | 52262  | 26131 | 2.000     | -0.583 | 0.646    |
| French 25 Size/BM, skipped-momentum panel     | Available | 0.500 | 45201  | 3477  | 13.000    | 0.667  | 3.000    |

Table 9. Row-naive and date-level p-value diagnostics for panel candidates.

| Candidate                                     | thr   | row t | HAC z | row p+   | HAC p+   | RW p     |
|-----------------------------------------------|-------|-------|-------|----------|----------|----------|
| French momentum deciles, rank-threshold panel | 0.700 | 3.584 | 3.790 | 1.70e-04 | 7.54e-05 | 2.00e-04 |
| French WML decile benchmark                   | 0.900 | 3.299 | 4.395 | 4.85e-04 | 5.53e-06 | 2.00e-04 |
| French 25 Size/BM, skipped-momentum panel     | 0.500 | 0.664 | 0.222 | 0.253    | 0.412    | 0.480    |

Table 10. Unclipped and one-sided floored UVIF diagnostics for panel candidates.

| Candidate                                     | rho    | SE infl. | UVIF unclipped | UVIF displayed | ERIR unclipped | ERIR displayed |
|-----------------------------------------------|--------|----------|----------------|----------------|----------------|----------------|
| French momentum deciles, rank-threshold panel | -0.138 | 0.766    | 0.587          | 1.000          | 1.704          | 1.000          |
| French WML decile benchmark                   | -0.583 | 0.646    | 0.417          | 1.000          | 2.399          | 1.000          |
| French 25 Size/BM, skipped-momentum panel     | 0.667  | 3.000    | 9.002          | 9.002          | 0.111          | 0.111          |

*Note:* Displayed main-text UVIF values are floored at one when the equicorrelation diagnostic is non-inflationary; unclipped values remain descriptive diagnostics and are not rejection rules.



Table 11. Sampling-boundary summary for panel candidates.

| Candidate                                                                                                                                                                                    | Row p+   | UVIF  | Max p+   | Net SR | Status                                             |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|-------|----------|--------|----------------------------------------------------|
| French momentum deciles, rank-threshold panel                                                                                                                                                | 1.70e-04 | 1.000 | 2.00e-04 | 0.447  | Permutation uninformative; remaining criteria pass |
| French WML decile benchmark                                                                                                                                                                  | 4.85e-04 | 1.000 | 2.00e-04 | 0.497  | Passes full evaluation                             |
| French 25 Size/BM, skipped-momentum panel                                                                                                                                                    | 0.253    | 9.002 | 0.943    | -0.050 | Fails statistical and cost checks                  |
| <i>Note:</i> Displayed UVIF is a one-sided inflation summary floored at one. Degenerate same-date permutation diagnostics are reported as uninformative rather than converted into failures. |          |       |          |        |                                                    |

Table 12. Horizon-effect sensitivity for the dynamic Size/BM momentum panel.

| L  | thr   | dates | avg names | gross SR | net SR 5bps | turnover | UVIF SR | serial UVIF | HAC p+ |
|----|-------|-------|-----------|----------|-------------|----------|---------|-------------|--------|
| 1  | 0.500 | 3497  | 12.913    | 0.073    | -0.394      | 0.733    | 11.498  | 0.891       | 0.386  |
| 5  | 0.500 | 3493  | 13.000    | -0.312   | -0.528      | 0.339    | 12.880  | 0.990       | 0.878  |
| 21 | 0.500 | 3477  | 13.000    | 0.058    | -0.050      | 0.169    | 12.121  | 0.933       | 0.412  |

Table 13. Primary-threshold dependence-aware Sharpe inference.

| Candidate                                     | thr   | Method   | SR    | SE    | CI low | CI high | p+       | n     |
|-----------------------------------------------|-------|----------|-------|-------|--------|---------|----------|-------|
| AQR Betting Against Beta, equity factors      | 0.000 | block    | 0.741 | 0.106 | 0.533  | 0.951   | 2.00e-04 | 25281 |
| AQR Betting Against Beta, equity factors      | 0.000 | HAC      | 0.741 | 0.103 | 0.540  | 0.943   | 3.08e-13 | 25281 |
| AQR Betting Against Beta, equity factors      | 0.000 | prewhite | 0.741 | 0.104 | 0.538  | 0.945   | 4.38e-13 | 25281 |
| AQR HML Devil                                 | 0.000 | block    | 0.336 | 0.114 | 0.105  | 0.552   | 0.002    | 26590 |
| AQR HML Devil                                 | 0.000 | HAC      | 0.336 | 0.126 | 0.089  | 0.583   | 0.004    | 26590 |
| AQR HML Devil                                 | 0.000 | prewhite | 0.336 | 0.111 | 0.119  | 0.553   | 0.001    | 26590 |
| AQR Quality Minus Junk                        | 0.000 | block    | 0.536 | 0.133 | 0.276  | 0.795   | 2.00e-04 | 17681 |
| AQR Quality Minus Junk                        | 0.000 | HAC      | 0.536 | 0.131 | 0.279  | 0.794   | 2.26e-05 | 17681 |
| AQR Quality Minus Junk                        | 0.000 | prewhite | 0.536 | 0.128 | 0.285  | 0.787   | 1.40e-05 | 17681 |
| AQR Value and Momentum Everywhere             | 0.000 | block    | 0.572 | 0.156 | 0.268  | 0.882   | 2.00e-04 | 651   |
| AQR Value and Momentum Everywhere             | 0.000 | HAC      | 0.573 | 0.159 | 0.261  | 0.885   | 1.62e-04 | 651   |
| AQR Value and Momentum Everywhere             | 0.000 | prewhite | 0.573 | 0.158 | 0.263  | 0.882   | 1.42e-04 | 651   |
| French momentum deciles, rank-threshold panel | 0.700 | block    | 0.447 | 0.114 | 0.222  | 0.667   | 2.00e-04 | 26131 |
| French momentum deciles, rank-threshold panel | 0.700 | HAC      | 0.447 | 0.118 | 0.216  | 0.679   | 7.54e-05 | 26131 |
| French momentum deciles, rank-threshold panel | 0.700 | prewhite | 0.447 | 0.114 | 0.224  | 0.670   | 4.18e-05 | 26131 |
| French WML decile benchmark                   | 0.900 | block    | 0.497 | 0.111 | 0.275  | 0.708   | 2.00e-04 | 26131 |
| French WML decile benchmark                   | 0.900 | HAC      | 0.497 | 0.113 | 0.275  | 0.718   | 5.53e-06 | 26131 |
| French WML decile benchmark                   | 0.900 | prewhite | 0.497 | 0.111 | 0.279  | 0.715   | 3.99e-06 | 26131 |
| French 25 Size/BM, skipped-momentum panel     | 0.500 | block    | 0.058 | 0.261 | -0.483 | 0.540   | 0.422    | 3477  |
| French 25 Size/BM, skipped-momentum panel     | 0.500 | HAC      | 0.058 | 0.260 | -0.452 | 0.567   | 0.412    | 3477  |
| French 25 Size/BM, skipped-momentum panel     | 0.500 | prewhite | 0.058 | 0.255 | -0.442 | 0.557   | 0.410    | 3477  |

Table 14. Same-date signal-permutation diagnostics for panel candidates.

| Candidate                                     | thr   | obs SR | null mean | null sd  | p+       | status                      | perms |
|-----------------------------------------------|-------|--------|-----------|----------|----------|-----------------------------|-------|
| French momentum deciles, rank-threshold panel | 0.000 | 0.395  | 0.410     | 0.039    | 0.646    | Available                   | 10000 |
| French momentum deciles, rank-threshold panel | 0.300 | 0.419  | 0.434     | 0.024    | 0.733    | Available                   | 10000 |
| French momentum deciles, rank-threshold panel | 0.500 | 0.419  | 0.434     | 0.024    | 0.733    | Available                   | 10000 |
| French momentum deciles, rank-threshold panel | 0.700 | 0.447  | 0.447     | 1.11e-16 | N/A      | Permutation null degenerate | 10000 |
| French momentum deciles, rank-threshold panel | 0.900 | 0.497  | 0.404     | 0.042    | 0.015    | Available                   | 10000 |
| French WML decile benchmark                   | 0.900 | 0.497  | -0.001    | 0.098    | 1.00e-04 | Available                   | 10000 |
| French 25 Size/BM, skipped-momentum panel     | 0.000 | 0.040  | 0.080     | 0.017    | 0.991    | Available                   | 10000 |
| French 25 Size/BM, skipped-momentum panel     | 0.300 | 0.034  | 0.097     | 0.023    | 0.997    | Available                   | 10000 |
| French 25 Size/BM, skipped-momentum panel     | 0.500 | 0.058  | 0.104     | 0.029    | 0.943    | Available                   | 10000 |
| French 25 Size/BM, skipped-momentum panel     | 0.700 | 0.087  | 0.137     | 0.041    | 0.892    | Available                   | 10000 |
| French 25 Size/BM, skipped-momentum panel     | 0.900 | 0.054  | 0.119     | 0.069    | 0.832    | Available                   | 10000 |

*Note:* A near-zero permutation-null standard deviation is reported as N/A because the placebo is uninformative for that threshold, not calibrated evidence for or against predictability.

Table 15. HAC bandwidth, prewhitening, and fixed-b robustness diagnostics.

| Candidate                                     | Diagnostic  | SR    | SE    | p+       |
|-----------------------------------------------|-------------|-------|-------|----------|
| AQR Betting Against Beta, equity factors      | HAC K=auto  | 0.741 | 0.103 | 3.08e-13 |
| AQR Betting Against Beta, equity factors      | HAC K=21    | 0.741 | 0.119 | 2.04e-10 |
| AQR Betting Against Beta, equity factors      | HAC K=63    | 0.741 | 0.130 | 6.06e-09 |
| AQR Betting Against Beta, equity factors      | prewhite    | 0.741 | 0.104 | 4.38e-13 |
| AQR Betting Against Beta, equity factors      | fixed-b 0.1 | 0.741 | 0.166 | 0.002    |
| AQR HML Devil                                 | HAC K=auto  | 0.336 | 0.126 | 0.004    |
| AQR HML Devil                                 | HAC K=21    | 0.336 | 0.128 | 0.004    |
| AQR HML Devil                                 | HAC K=63    | 0.336 | 0.137 | 0.007    |
| AQR HML Devil                                 | prewhite    | 0.336 | 0.111 | 0.001    |
| AQR HML Devil                                 | fixed-b 0.1 | 0.336 | 0.110 | 0.004    |
| AQR Quality Minus Junk                        | HAC K=auto  | 0.536 | 0.131 | 2.26e-05 |
| AQR Quality Minus Junk                        | HAC K=21    | 0.536 | 0.140 | 6.33e-05 |
| AQR Quality Minus Junk                        | HAC K=63    | 0.536 | 0.147 | 1.32e-04 |
| AQR Quality Minus Junk                        | prewhite    | 0.536 | 0.128 | 1.40e-05 |
| AQR Quality Minus Junk                        | fixed-b 0.1 | 0.536 | 0.151 | 0.002    |
| AQR Value and Momentum Everywhere             | HAC K=auto  | 0.573 | 0.159 | 1.62e-04 |
| AQR Value and Momentum Everywhere             | HAC K=21    | 0.573 | 0.155 | 1.14e-04 |
| AQR Value and Momentum Everywhere             | HAC K=63    | 0.573 | 0.144 | 3.41e-05 |
| AQR Value and Momentum Everywhere             | prewhite    | 0.573 | 0.158 | 1.42e-04 |
| AQR Value and Momentum Everywhere             | fixed-b 0.1 | 0.573 | 0.143 | 0.004    |
| French momentum deciles, rank-threshold panel | HAC K=auto  | 0.447 | 0.118 | 7.54e-05 |
| French momentum deciles, rank-threshold panel | HAC K=21    | 0.447 | 0.120 | 9.94e-05 |
| French momentum deciles, rank-threshold panel | HAC K=63    | 0.447 | 0.122 | 1.21e-04 |
| French momentum deciles, rank-threshold panel | prewhite    | 0.447 | 0.114 | 4.18e-05 |
| French momentum deciles, rank-threshold panel | fixed-b 0.1 | 0.447 | 0.143 | 0.004    |
| French WML decile benchmark                   | HAC K=auto  | 0.497 | 0.113 | 5.53e-06 |
| French WML decile benchmark                   | HAC K=21    | 0.497 | 0.116 | 9.62e-06 |
| French WML decile benchmark                   | HAC K=63    | 0.497 | 0.117 | 1.01e-05 |
| French WML decile benchmark                   | prewhite    | 0.497 | 0.111 | 3.99e-06 |
| French WML decile benchmark                   | fixed-b 0.1 | 0.497 | 0.147 | 0.002    |
| French 25 Size/BM, skipped-momentum panel     | HAC K=auto  | 0.058 | 0.260 | 0.412    |
| French 25 Size/BM, skipped-momentum panel     | HAC K=21    | 0.058 | 0.249 | 0.409    |
| French 25 Size/BM, skipped-momentum panel     | HAC K=63    | 0.058 | 0.231 | 0.401    |
| French 25 Size/BM, skipped-momentum panel     | prewhite    | 0.058 | 0.255 | 0.410    |
| French 25 Size/BM, skipped-momentum panel     | fixed-b 0.1 | 0.058 | 0.172 | 0.360    |

Table 16. Decision-rule sensitivity by alpha threshold.

| alpha | candidates | full passes | applicable passes |
|-------|------------|-------------|-------------------|
| 0.010 | 8          | 1           | 6                 |
| 0.050 | 8          | 1           | 6                 |

Table 17. Holdout and subperiod Sharpe diagnostics.

| Candidate                                     | window        | SR     | n     |
|-----------------------------------------------|---------------|--------|-------|
| AQR Betting Against Beta, equity factors      | train_70pct   | 0.887  | 17696 |
| AQR Betting Against Beta, equity factors      | holdout_30pct | 0.531  | 7585  |
| AQR Betting Against Beta, equity factors      | quarter_min   | 0.280  | 6320  |
| AQR Betting Against Beta, equity factors      | quarter_max   | 1.270  | 6320  |
| AQR HML Devil                                 | train_70pct   | 0.465  | 18613 |
| AQR HML Devil                                 | holdout_30pct | 0.120  | 7977  |
| AQR HML Devil                                 | quarter_min   | 0.273  | 6647  |
| AQR HML Devil                                 | quarter_max   | 0.445  | 6647  |
| AQR Quality Minus Junk                        | train_70pct   | 0.794  | 12376 |
| AQR Quality Minus Junk                        | holdout_30pct | 0.305  | 5305  |
| AQR Quality Minus Junk                        | quarter_min   | 0.174  | 4421  |
| AQR Quality Minus Junk                        | quarter_max   | 1.083  | 4420  |
| AQR Value and Momentum Everywhere             | train_70pct   | 0.626  | 455   |
| AQR Value and Momentum Everywhere             | holdout_30pct | 0.431  | 196   |
| AQR Value and Momentum Everywhere             | quarter_min   | 0.372  | 163   |
| AQR Value and Momentum Everywhere             | quarter_max   | 0.941  | 163   |
| French momentum deciles, rank-threshold panel | train_70pct   | 0.615  | 18292 |
| French momentum deciles, rank-threshold panel | holdout_30pct | 0.229  | 7839  |
| French momentum deciles, rank-threshold panel | quarter_min   | 0.171  | 6532  |
| French momentum deciles, rank-threshold panel | quarter_max   | 1.280  | 6533  |
| French WML decile benchmark                   | train_70pct   | 0.641  | 18292 |
| French WML decile benchmark                   | holdout_30pct | 0.310  | 7839  |
| French WML decile benchmark                   | quarter_min   | 0.224  | 6533  |
| French WML decile benchmark                   | quarter_max   | 1.370  | 6533  |
| French 25 Size/BM, skipped-momentum panel     | train_70pct   | 0.088  | 2428  |
| French 25 Size/BM, skipped-momentum panel     | holdout_30pct | -0.007 | 1049  |
| French 25 Size/BM, skipped-momentum panel     | quarter_min   | -0.581 | 853   |
| French 25 Size/BM, skipped-momentum panel     | quarter_max   | 0.210  | 875   |

Table 18. Turnover-scaled cost sensitivity and break-even cost.

| Candidate                                     | cost bps | turnover | gross SR | net SR | break-even bps |
|-----------------------------------------------|----------|----------|----------|--------|----------------|
| AQR Betting Against Beta, equity factors      | 0        | 0.000    | 0.741    | 0.741  |                |
| AQR Betting Against Beta, equity factors      | 1        | 0.000    | 0.741    | 0.741  |                |
| AQR Betting Against Beta, equity factors      | 5        | 0.000    | 0.741    | 0.741  |                |
| AQR Betting Against Beta, equity factors      | 10       | 0.000    | 0.741    | 0.741  |                |
| AQR Betting Against Beta, equity factors      | 25       | 0.000    | 0.741    | 0.741  |                |
| AQR HML Devil                                 | 0        | 0.000    | 0.336    | 0.336  |                |
| AQR HML Devil                                 | 1        | 0.000    | 0.336    | 0.336  |                |
| AQR HML Devil                                 | 5        | 0.000    | 0.336    | 0.336  |                |
| AQR HML Devil                                 | 10       | 0.000    | 0.336    | 0.336  |                |
| AQR HML Devil                                 | 25       | 0.000    | 0.336    | 0.336  |                |
| AQR Quality Minus Junk                        | 0        | 0.000    | 0.536    | 0.536  |                |
| AQR Quality Minus Junk                        | 1        | 0.000    | 0.536    | 0.536  |                |
| AQR Quality Minus Junk                        | 5        | 0.000    | 0.536    | 0.536  |                |
| AQR Quality Minus Junk                        | 10       | 0.000    | 0.536    | 0.536  |                |
| AQR Quality Minus Junk                        | 25       | 0.000    | 0.536    | 0.536  |                |
| AQR Value and Momentum Everywhere             | 0        | 0.000    | 0.572    | 0.572  |                |
| AQR Value and Momentum Everywhere             | 1        | 0.000    | 0.572    | 0.572  |                |
| AQR Value and Momentum Everywhere             | 5        | 0.000    | 0.572    | 0.572  |                |
| AQR Value and Momentum Everywhere             | 10       | 0.000    | 0.572    | 0.572  |                |
| AQR Value and Momentum Everywhere             | 25       | 0.000    | 0.572    | 0.572  |                |
| French momentum deciles, rank-threshold panel | 0        | 1.91e-05 | 0.447    | 0.447  | 85893.000      |
| French momentum deciles, rank-threshold panel | 1        | 1.91e-05 | 0.447    | 0.447  | 85893.000      |
| French momentum deciles, rank-threshold panel | 5        | 1.91e-05 | 0.447    | 0.447  | 85893.000      |
| French momentum deciles, rank-threshold panel | 10       | 1.91e-05 | 0.447    | 0.447  | 85893.000      |
| French momentum deciles, rank-threshold panel | 25       | 1.91e-05 | 0.447    | 0.447  | 85893.000      |
| French WML decile benchmark                   | 0        | 1.91e-05 | 0.497    | 0.497  | 122232.000     |
| French WML decile benchmark                   | 1        | 1.91e-05 | 0.497    | 0.497  | 122232.000     |
| French WML decile benchmark                   | 5        | 1.91e-05 | 0.497    | 0.497  | 122232.000     |
| French WML decile benchmark                   | 10       | 1.91e-05 | 0.497    | 0.497  | 122232.000     |
| French WML decile benchmark                   | 25       | 1.91e-05 | 0.497    | 0.497  | 122232.000     |
| French 25 Size/BM, skipped-momentum panel     | 0        | 0.169    | 0.058    | 0.058  | 2.683          |
| French 25 Size/BM, skipped-momentum panel     | 1        | 0.169    | 0.058    | 0.036  | 2.683          |
| French 25 Size/BM, skipped-momentum panel     | 5        | 0.169    | 0.058    | -0.050 | 2.683          |
| French 25 Size/BM, skipped-momentum panel     | 10       | 0.169    | 0.058    | -0.157 | 2.683          |
| French 25 Size/BM, skipped-momentum panel     | 25       | 0.169    | 0.058    | -0.480 | 2.683          |

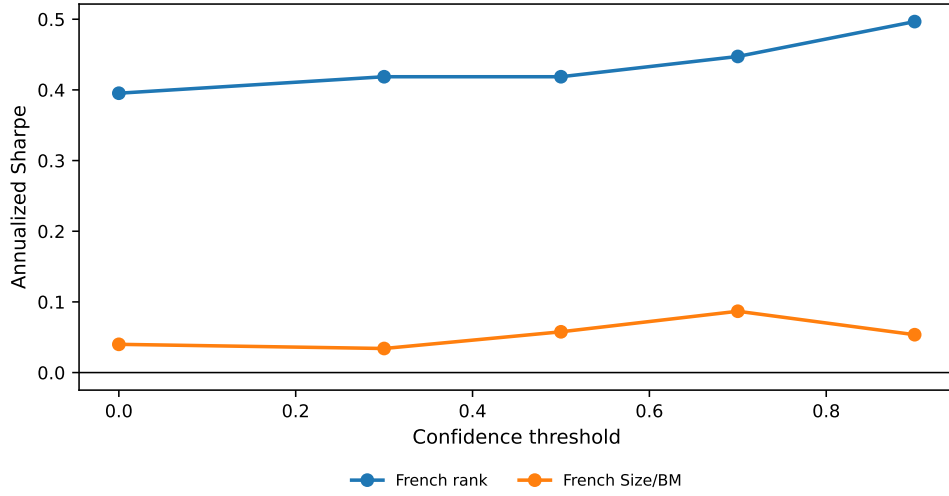


Figure 1. Threshold-profile diagnostics for public candidate panels; each line shows how annualized Sharpe changes when the pre-specified confidence threshold is varied.

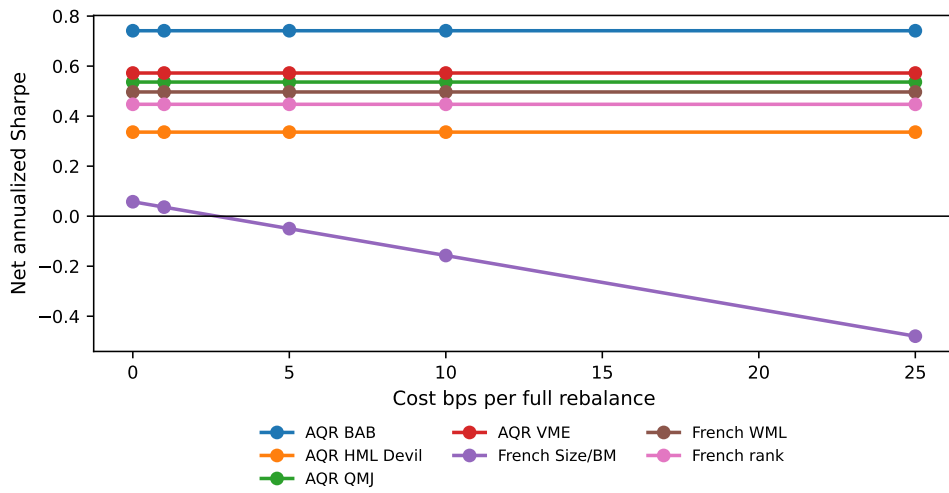


Figure 2. Turnover-scaled cost sensitivity at primary candidate thresholds; positive values indicate that the candidate remains above zero after the displayed linear cost stress.

## 9 Full Monte Carlo Tables

The simulation tables target the date-level Sharpe claim. Row-naive coverage is therefore reported against that date-level target, not against a separate row-distribution target. The design-sweep table is a stress diagnostic over same-date correlation and selected-count pressure under a zero-edge null. The main concern in this table group is size distortion: a method that looks powerful because it overrejects under the null is not acceptable for confirmatory evaluation.

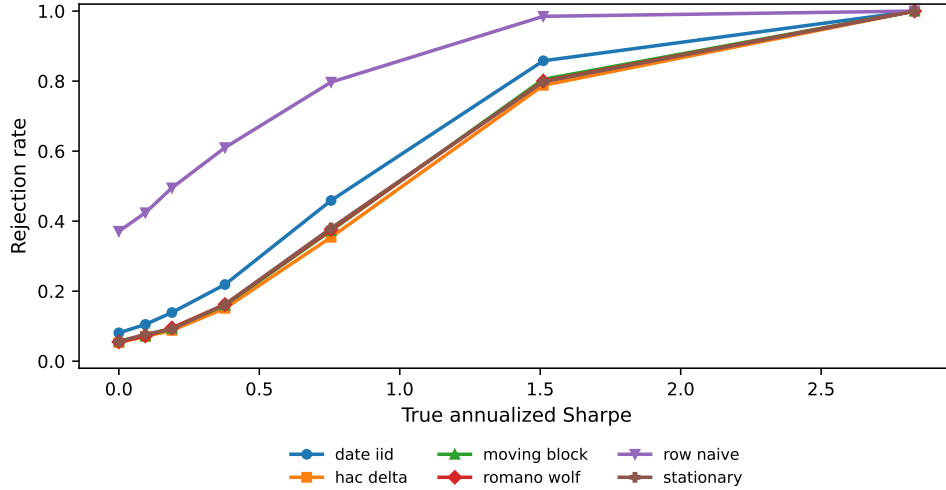


Figure 3. Monte Carlo rejection rates by true annualized Sharpe; the figure separates genuine power from methods that already overreject under zero-edge null designs.

Table 19. Monte Carlo design contract.

| Design           | T    | N   | Selected count       | XS dep.             | Serial dep.  | Heterosk.        | Reps | Resampling | Seed      |
|------------------|------|-----|----------------------|---------------------|--------------|------------------|------|------------|-----------|
| IID null         | 1000 | 50  | all rows             | $\rho=0$            | $AR(1)=0$    | none             | 1000 | 5000 boot  | rep index |
| Dependent null   | 1000 | 50  | all rows             | $\rho=0.2$          | $AR(1)=0.2$  | none             | 1000 | 5000 boot  | rep index |
| GARCH null       | 1000 | 50  | all rows             | $\rho=0.2$          | $AR(1)=0.2$  | GARCH(0.05,0.90) | 1000 | 5000 boot  | rep index |
| Positive control | 5000 | 100 | signal-selected rows | common loading=0.50 | $AR(1)=0.10$ | none             | 100  | 499 boot   | 777       |

*Note:* All null designs have true Sharpe equal to zero. Positive-control rows use a declared nonzero signal and are included to show that the reporting rule is not mechanically anti-discovery.

## 10 Robustness and Configuration Notes

The public configuration is fixed in `code/campaign_registry.json`. It records loaders, public source metadata, threshold menus, primary thresholds, and lookback/skip parameters. Changing a source, lookback, threshold, cost assumption, or eligibility rule after seeing results creates a new researcher menu and must be reported as exploratory. Rendered tables in the submitted package use the following SHA256 identifiers:



Table 20. Monte Carlo and resampling settings.

| Setting                          | Value                                                             |
|----------------------------------|-------------------------------------------------------------------|
| Monte Carlo replications         | 1000                                                              |
| Bootstrap draws                  | 5000                                                              |
| Permutation draws                | 10000                                                             |
| Main null panel sizes            | T=1000, N=50                                                      |
| Cross-sectional correlation grid | 0.000, 0.200, 0.250, 0.300, 0.350, 0.400, 0.500                   |
| AR(1) grid                       | 0.000, 0.150, 0.200, 0.300, 0.500, 0.700                          |
| GARCH parameter pairs            | (0.050, 0.900); (0.100, 0.850); (0.150, 0.800)                    |
| HAC bandwidth rule               | automatic Bartlett, with fixed-lag sensitivity checks             |
| Moving-block length              | ceil(T to the 1/5 times max(D,1)), capped at floor(T/3)           |
| Stationary mean block            | same length as moving-block rule                                  |
| Dependence factor D              | (2 rho1/(1-rho1)) to the 2/3 for positive AR(1) rho1; otherwise 1 |
| Seed policy                      | seed-indexed replications from campaign runner                    |

Table 21. Simulation DGP parameter grid.

| DGP              | Name                   | T    | N   | mu       | sigma | AR1   | rho   | GARCH a | GARCH b | K | regime |
|------------------|------------------------|------|-----|----------|-------|-------|-------|---------|---------|---|--------|
| 01_iid           | IID baseline           | 1000 | 50  | 4.00e-04 | 0.015 | 0.000 | 0.000 | 0.000   | 0.000   | 1 | no     |
| 02_ser_mild      | Serial mild            | 1000 | 50  | 4.00e-04 | 0.015 | 0.200 | 0.000 | 0.000   | 0.000   | 1 | no     |
| 03_ser_strong    | Serial strong          | 1000 | 50  | 4.00e-04 | 0.015 | 0.500 | 0.000 | 0.000   | 0.000   | 1 | no     |
| 04_xs_mild       | Cross mild             | 1000 | 50  | 4.00e-04 | 0.015 | 0.000 | 0.200 | 0.000   | 0.000   | 1 | no     |
| 05_xs_strong     | Cross strong           | 1000 | 50  | 4.00e-04 | 0.015 | 0.000 | 0.500 | 0.000   | 0.000   | 1 | no     |
| 06_both_mod      | Both moderate          | 1000 | 50  | 4.00e-04 | 0.015 | 0.200 | 0.200 | 0.000   | 0.000   | 1 | no     |
| 07_both_strong   | Both strong            | 1000 | 50  | 4.00e-04 | 0.015 | 0.500 | 0.500 | 0.000   | 0.000   | 1 | no     |
| 08_realistic     | Realistic              | 2500 | 100 | 4.00e-04 | 0.015 | 0.150 | 0.350 | 0.000   | 0.000   | 1 | no     |
| 09_garch_mild    | GARCH mild             | 1000 | 50  | 4.00e-04 | 0.015 | 0.200 | 0.200 | 0.050   | 0.900   | 1 | no     |
| 10_garch_strong  | GARCH strong           | 1000 | 50  | 4.00e-04 | 0.015 | 0.300 | 0.400 | 0.100   | 0.850   | 1 | no     |
| 11_multifactor   | Multi-factor K=3       | 1000 | 50  | 4.00e-04 | 0.015 | 0.200 | 0.300 | 0.000   | 0.000   | 3 | no     |
| 12_regime        | Regime-switching       | 2000 | 50  | 4.00e-04 | 0.015 | 0.150 | 0.250 | 0.000   | 0.000   | 1 | yes    |
| 13_high_ar1      | High-persistence AR(1) | 1000 | 50  | 4.00e-04 | 0.015 | 0.700 | 0.300 | 0.000   | 0.000   | 1 | no     |
| 14_garch_highvol | High-volatility GARCH  | 1000 | 50  | 4.00e-04 | 0.015 | 0.200 | 0.300 | 0.150   | 0.800   | 1 | no     |

Table 22. Coverage of the date-level Sharpe target by alternative interval procedures.

| DGP              | blocked | hac_delta | iid   | row_naive | stationary |
|------------------|---------|-----------|-------|-----------|------------|
| 01_iid           | 0.947   | 0.952     | 0.951 | 0.000     | 0.947      |
| 02_ser_mild      | 0.938   | 0.943     | 0.898 | 0.000     | 0.936      |
| 03_ser_strong    | 0.921   | 0.934     | 0.734 | 0.000     | 0.919      |
| 04_xs_mild       | 0.950   | 0.950     | 0.952 | 0.073     | 0.950      |
| 05_xs_strong     | 0.934   | 0.938     | 0.942 | 0.262     | 0.938      |
| 06_both_mod      | 0.937   | 0.939     | 0.898 | 0.088     | 0.936      |
| 07_both_strong   | 0.934   | 0.943     | 0.771 | 0.178     | 0.935      |
| 08_realistic     | 0.937   | 0.941     | 0.901 | 0.107     | 0.937      |
| 09_garch_mild    | 0.926   | 0.940     | 0.877 | 0.110     | 0.930      |
| 10_garch_strong  | 0.926   | 0.948     | 0.862 | 0.199     | 0.927      |
| 11_multifactor   | 0.937   | 0.947     | 0.895 | 0.225     | 0.940      |
| 12_regime        | 0.929   | 0.934     | 0.890 | 0.306     | 0.929      |
| 13_high_ar1      | 0.899   | 0.933     | 0.562 | 0.137     | 0.907      |
| 14_garch_highvol | 0.940   | 0.948     | 0.895 | 0.193     | 0.941      |

*Note:* Row-naive intervals are evaluated against the economically relevant date-level Sharpe target, not against their own row-distribution Sharpe target. Low row-naive coverage therefore records target mismatch and false precision at the portfolio boundary.

Table 23. Row-naive interval behavior when evaluated against the date-level Sharpe target.

| DGP              | Name                   | row coverage | bias vs date SR | CI width | mean SE | n    |
|------------------|------------------------|--------------|-----------------|----------|---------|------|
| 01_iid           | IID baseline           | 0.000        | -2.567          | 0.278    | 0.071   | 1000 |
| 02_ser_mild      | Serial mild            | 0.000        | -2.567          | 0.278    | 0.071   | 1000 |
| 03_ser_strong    | Serial strong          | 0.000        | -2.568          | 0.278    | 0.071   | 1000 |
| 04_xs_mild       | Cross mild             | 0.073        | -0.474          | 0.278    | 0.071   | 1000 |
| 05_xs_strong     | Cross strong           | 0.262        | -0.153          | 0.278    | 0.071   | 1000 |
| 06_both_mod      | Both moderate          | 0.088        | -0.503          | 0.278    | 0.071   | 1000 |
| 07_both_strong   | Both strong            | 0.178        | -0.145          | 0.278    | 0.071   | 1000 |
| 08_realistic     | Realistic              | 0.107        | -0.277          | 0.124    | 0.032   | 1000 |
| 09_garch_mild    | GARCH mild             | 0.110        | -0.473          | 0.278    | 0.071   | 1000 |
| 10_garch_strong  | GARCH strong           | 0.199        | -0.268          | 0.278    | 0.071   | 1000 |
| 11_multifactor   | Multi-factor K=3       | 0.225        | -0.327          | 0.278    | 0.071   | 1000 |
| 12_regime        | Regime-switching       | 0.306        | -0.092          | 0.197    | 0.050   | 1000 |
| 13_high_ar1      | High-persistence AR(1) | 0.137        | -0.338          | 0.278    | 0.071   | 1000 |
| 14_garch_highvol | High-volatility GARCH  | 0.193        | -0.325          | 0.278    | 0.071   | 1000 |

*Note:* These are not ordinary row-estimator coverage rates; they evaluate the row-naive interval against the economically relevant date-level target.

Table 24. Sampling-boundary stress tests under a zero-edge null.

| Panel                           | Setting   | Row rej. | HAC rej. | Row mean SE | HAC mean SE |
|---------------------------------|-----------|----------|----------|-------------|-------------|
| A. Same-date correlation (N=50) | rho=0.000 | 0.097    | 0.056    | 0.071       | 0.598       |
| A. Same-date correlation (N=50) | rho=0.100 | 0.310    | 0.064    | 0.071       | 0.598       |
| A. Same-date correlation (N=50) | rho=0.300 | 0.354    | 0.044    | 0.071       | 0.599       |
| A. Same-date correlation (N=50) | rho=0.350 | 0.368    | 0.071    | 0.071       | 0.597       |
| A. Same-date correlation (N=50) | rho=0.500 | 0.409    | 0.047    | 0.071       | 0.597       |
| A. Same-date correlation (N=50) | rho=0.700 | 0.408    | 0.060    | 0.071       | 0.599       |
| B. Selected count (rho=0.35)    | N=5       | 0.192    | 0.058    | 0.225       | 0.598       |
| B. Selected count (rho=0.35)    | N=25      | 0.308    | 0.052    | 0.100       | 0.600       |
| B. Selected count (rho=0.35)    | N=50      | 0.368    | 0.071    | 0.071       | 0.597       |
| B. Selected count (rho=0.35)    | N=100     | 0.421    | 0.056    | 0.050       | 0.598       |

*Note:* Row-naive overrejection rises with same-date dependence and selected-count pressure, whereas date-level HAC remains close to nominal size.

Table 25. Null rejection rates at nominal 5 percent size.

| dgp        | date_iid | hac_delta | moving_block | romano_wolf | row_naive | stationary |
|------------|----------|-----------|--------------|-------------|-----------|------------|
| null_dep   | 0.094    | 0.066     | 0.070        | 0.068       | 0.352     | 0.067      |
| null_garch | 0.098    | 0.063     | 0.070        | 0.067       | 0.351     | 0.068      |
| null_iid   | 0.060    | 0.060     | 0.061        | 0.060       | 0.062     | 0.062      |

Table 26. Rejection rates across true annualized Sharpe values.

| index | true_sharpe | date_iid | hac_delta | moving_block | romano_wolf | row_naive | stationary |
|-------|-------------|----------|-----------|--------------|-------------|-----------|------------|
| 0.000 | 0.000       | 0.081    | 0.053     | 0.058        | 0.055       | 0.371     | 0.056      |
| 1.000 | 0.094       | 0.105    | 0.071     | 0.072        | 0.072       | 0.424     | 0.077      |
| 2.000 | 0.189       | 0.139    | 0.088     | 0.094        | 0.095       | 0.495     | 0.092      |
| 3.000 | 0.378       | 0.219    | 0.151     | 0.160        | 0.162       | 0.610     | 0.162      |
| 4.000 | 0.755       | 0.459    | 0.354     | 0.373        | 0.376       | 0.797     | 0.380      |
| 5.000 | 1.511       | 0.858    | 0.788     | 0.805        | 0.800       | 0.985     | 0.798      |
| 6.000 | 2.833       | 1.000    | 1.000     | 1.000        | 1.000       | 1.000     | 1.000      |

```
configuration code/campaign_registry.json:  
3a616f119f3fe2603b0663e3f68c499496c37a1781f3003e82814c9269a16259  
generated campaign metadata campaign_metadata.json:  
5cb21ac2cb87cb74a26090a66b28dac6890f76dd1d4bd4391ea6290267810dff
```

Stationarity diagnostics, holdout splits, HAC bandwidth sweeps, prewhitened HAC, fixed-fraction HAC, moving-block bootstrap, and stationary bootstrap results are secondary robustness checks. They are used to identify fragile claims, not to replace the main date-return decision rule.